

Benchmark comparison tables — acquisition $\times$ surrogate model

Legend. Each cell is the final metric (mean $\pm$ sd over seeds). **Bold** marks the best model for that acquisition; **green** marks the single best (acquisition, model) combination for that problem. Lower is better for both metrics.

Table 1: Final regret (value $- f^*$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 10-D	Ackley 2-D	Branin	Golinski	Griewank 2-D	
LCB	Standard LVGP	–	0.00641 $\pm$ 0.0063	0.0911 $\pm$ 0.057	$2.56 \times 10^{-6} \pm 8.5 \times 10^{-8}$	0.00413 $\pm$ 0.015	1.53×10^{-7}
	Heteroscedastic LVGP	–	0.0302 $\pm$ 0.035	0.0913 $\pm$ 0.057	$1.24 \times 10^{-7} \pm 1 \times 10^{-8}$	0.00858 $\pm$ 0.021	4.33 $\times 10^{-7}$
	Separate GP	–	0.624 $\pm$ 1.2	0.649 $\pm$ 1.1	0 $\pm$ 0	0.0912 $\pm$ 0.1	0
	Categorical kernel	–	0.196 $\pm$ 0.71	0.0516 $\pm$ 0.088	0 $\pm$ 0	0.00441 $\pm$ 0.015	3.07×10^{-7}
PI	Standard LVGP	–	0.0127 $\pm$ 0.015	0.0697 $\pm$ 0.057	31.6 $\pm$ 17	0.000736 $\pm$ 0.0011	0
	Heteroscedastic LVGP	–	0.547 $\pm$ 1.2	0.289 $\pm$ 0.91	0.941 $\pm$ 1.9	0.0395 $\pm$ 0.072	0
	Separate GP	–	0.12 $\pm$ 0.55	0.179 $\pm$ 0.42	0 $\pm$ 0	0.026 $\pm$ 0.048	1.74 $\times 10^{-7}$
	Categorical kernel	–	0.0145 $\pm$ 0.012	0.0736 $\pm$ 0.088	0 $\pm$ 0	0.000795 $\pm$ 0.0019	0
EI	Standard LVGP	–	0.0123 $\pm$ 0.012	0.0532 $\pm$ 0.054	16.3 $\pm$ 17	3.11 $\times 10^{-7} \pm 3.7 \times 10^{-6}$	0
	Heteroscedastic LVGP	–	0.0212 $\pm$ 0.024	0.092 $\pm$ 0.049	0.000119 $\pm$ 0.00015	0.00697 $\pm$ 0.019	0
	Separate GP	–	0.221 $\pm$ 0.76	0.419 $\pm$ 0.71	0 $\pm$ 0	0.0292 $\pm$ 0.049	0.000119
	Categorical kernel	–	0.0132 $\pm$ 0.014	0.0898 $\pm$ 0.092	0 $\pm$ 0	0.000435 $\pm$ 0.00059	0
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	–	0.0166 $\pm$ 0.016	0.0862 $\pm$ 0.054	3.75 $\pm$ 2.1	0.00464 $\pm$ 0.016	0
	Separate GP	–	0.217 $\pm$ 0.77	0.0966 $\pm$ 0.12	0 $\pm$ 0	0.0271 $\pm$ 0.049	0.000119
	Categorical kernel	–	0.0122 $\pm$ 0.012	0.0842 $\pm$ 0.088	0 $\pm$ 0	0.00088 $\pm$ 0.0015	0
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	–	0.024 $\pm$ 0.025	0.078 $\pm$ 0.085	0.307 $\pm$ 0.61	0.0137 $\pm$ 0.024	0
	Separate GP	–	0.027 $\pm$ 0.027	0.19 $\pm$ 0.39	0 $\pm$ 0	0.0467 $\pm$ 0.061	0.000119
	Categorical kernel	–	0.0191 $\pm$ 0.024	0.131 $\pm$ 0.41	1.49 $\pm$ 3	0.0267 $\pm$ 0.029	0
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	–	0.0264 $\pm$ 0.024	0.00249 $\pm$ 0.0073	9.65 $\pm$ 6.6	0.00853 $\pm$ 0.021	6.33 $\times 10^{-7}$
	Separate GP	–	0.723 $\pm$ 1.3	0.377 $\pm$ 0.91	56.7 $\pm$ 47	0.1 $\pm$ 0.12	0
	Categorical kernel	–	0.11 $\pm$ 0.55	0.0269 $\pm$ 0.081	20 $\pm$ 29	0.00463 $\pm$ 0.015	1.23×10^{-7}

Table 2: Final regret (value $-f^*$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 10-D	Ackley 2-D	Branin	Golinski	Griewank 2-D	
LCB	Standard LVGP	1.13 ± 0.23	0.00408 ± 0.0047	0.00124 ± 0.0059	$2.5 \times 10^{-6} \pm 8.5 \times 10^{-8}$	0.00413 ± 0.015	
	Heteroscedastic LVGP	0.655 ± 0.12	0.0188 ± 0.021	0.0993 ± 0.063	$1.26 \times 10^{-7} \pm 1.3 \times 10^{-9}$	0.0105 ± 0.023	
	Separate GP	1.39 ± 0.68	0.726 ± 1.3	0.529 ± 0.88	0 ± 0	0.0769 ± 0.094	
	Categorical kernel	1.17 ± 0.23	0.294 ± 0.88	0.0941 ± 0.12	0 ± 0	0.0183 ± 0.06	
PI	Standard LVGP	0.477 ± 0.32	0.0131 ± 0.014	0.00462 ± 0.023	43.8 ± 0	0.0009 ± 0.0012	
	Heteroscedastic LVGP	–	0.28 ± 1	0.238 ± 0.59	1.37 ± 2.6	0.0426 ± 0.081	
	Separate GP	1.75 ± 0.94	0.709 ± 1.3	0.238 ± 0.59	0 ± 0	0.0349 ± 0.062	6
	Categorical kernel	$4.44 \times 10^{-16} \pm 0$	0.00491 ± 0.006	0.0548 ± 0.066	0 ± 0	0.0025 ± 0.011	
EI	Standard LVGP	1.47 ± 0.73	0.0102 ± 0.0098	0.00181 ± 0.0076	37.3 ± 13	$-6.54 \times 10^{-7} \pm 4.2 \times 10^{-7}$	
	Heteroscedastic LVGP	–	0.022 ± 0.031	0.0802 ± 0.058	$6.39 \times 10^{-5} \pm 7.9 \times 10^{-5}$	0.0225 ± 0.048	
	Separate GP	1.75 ± 0.93	0.422 ± 1	0.662 ± 0.95	0 ± 0	0.029 ± 0.049	3.0
	Categorical kernel	$4.44 \times 10^{-16} \pm 0$	0.01 ± 0.0092	0.158 ± 0.39	0 ± 0	0.000174 ± 0.00025	
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	–	0.0259 ± 0.023	0.0211 ± 0.035	7.82 ± 4.1	0.00868 ± 0.021	
	Separate GP	2.4 ± 0.41	0.425 ± 1	0.434 ± 0.74	0 ± 0	0.0288 ± 0.049	3.0
	Categorical kernel	$4.44 \times 10^{-16} \pm 0$	0.00801 ± 0.0085	0.0421 ± 0.048	0 ± 0	0.0095 ± 0.021	
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	–	0.0208 ± 0.019	0.00541 ± 0.014	0.965 ± 1.4	0.0194 ± 0.028	
	Separate GP	3.13 ± 1.2	0.116 ± 0.55	0.666 ± 1.1	0 ± 0	0.0413 ± 0.048	
	Categorical kernel	2.05 ± 0.32	0.00552 ± 0.0094	0.135 ± 0.41	0 ± 0	0.0406 ± 0.049	
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	1.28 ± 0	0.0203 ± 0.025	0.00189 ± 0.0049	2.44 ± 2.5	0.0085 ± 0.021	
	Separate GP	1.72 ± 1.1	0.723 ± 1.3	0.288 ± 0.78	17.5 ± 28	0.0768 ± 0.094	
	Categorical kernel	0.579 ± 0.38	0.397 ± 1	0.154 ± 0.57	10.6 ± 4	0.0411 ± 0.1	

Table 3: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 10-D	Ackley 2-D	Branin	Golinski	Griewank 2-D	OTL Circuit
LCB	Standard LVGP	–	$0.01 \pm 5.3 \times 10^{-8}$	23.8 ± 22	$576 \pm 3.9 \times 10^{-8}$	0.00608 ± 0.0012	$0.000194 \pm 5.3 \times 10^{-11}$
	Heteroscedastic LVGP	–	$0.01 \pm 9 \times 10^{-7}$	26.6 ± 22	$576 \pm 4.9 \times 10^{-9}$	0.00576 ± 0.0016	$0.000194 \pm 1.7 \times 10^{-11}$
	Separate GP	–	0.016 ± 0.012	23.7 ± 27	576 ± 0	0.00892 ± 0.0073	0.000651 ± 0.00091
	Categorical kernel	–	0.0111 ± 0.0054	10.8 ± 19	576 ± 0	0.00608 ± 0.0012	0.000194 ± 0
PI	Standard LVGP	–	$0.01 \pm 2.2 \times 10^{-7}$	12 ± 19	597 ± 19	$0.0064 \pm 2.3 \times 10^{-6}$	0.000753 ± 0.00028
	Heteroscedastic LVGP	–	0.0118 ± 0.0059	24.7 ± 24	$576 \pm 4 \times 10^{-5}$	0.0066 ± 0.0048	0.000514 ± 0.00032
	Separate GP	–	0.011 ± 0.0054	12.9 ± 19	576 ± 0	0.00556 ± 0.0034	0.000194 ± 0
	Categorical kernel	–	$0.01 \pm 1.7 \times 10^{-7}$	15.9 ± 21	576 ± 0	$0.0064 \pm 3.9 \times 10^{-6}$	0.000908 ± 0.0008
EI	Standard LVGP	–	$0.01 \pm 1.5 \times 10^{-7}$	10.4 ± 18	584 ± 11	$0.0064 \pm 7.5 \times 10^{-9}$	0.00079 ± 0.0002
	Heteroscedastic LVGP	–	$0.01 \pm 5.4 \times 10^{-7}$	23.4 ± 22	$576 \pm 4 \times 10^{-6}$	0.00592 ± 0.0014	0.000514 ± 0.00032
	Separate GP	–	0.012 ± 0.0075	14.2 ± 18	576 ± 0	0.00524 ± 0.0035	$0.000275 \pm 9.9 \times 10^{-5}$
	Categorical kernel	–	$0.01 \pm 1.8 \times 10^{-7}$	17.3 ± 21	576 ± 0	$0.0064 \pm 1.2 \times 10^{-6}$	0.000514 ± 0.00032
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	–	$0.01 \pm 2.7 \times 10^{-7}$	25 ± 22	$576 \pm 9.1 \times 10^{-6}$	0.00608 ± 0.0012	0.000512 ± 0.00032
	Separate GP	–	0.012 ± 0.0075	16.4 ± 21	576 ± 0	0.0054 ± 0.0034	$0.000235 \pm 8.1 \times 10^{-5}$
	Categorical kernel	–	$0.01 \pm 1.5 \times 10^{-7}$	14.9 ± 21	576 ± 0	$0.0064 \pm 3.1 \times 10^{-6}$	0.000275 ± 0.0001
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	–	$0.01 \pm 4.8 \times 10^{-7}$	15.1 ± 21	576 ± 0.00033	0.00544 ± 0.0019	$0.000275 \pm 9.9 \times 10^{-5}$
	Separate GP	–	$0.01 \pm 6.2 \times 10^{-7}$	13.6 ± 21	576 ± 0	0.00505 ± 0.0045	$0.000275 \pm 9.9 \times 10^{-5}$
	Categorical kernel	–	$0.01 \pm 6 \times 10^{-7}$	9.12 ± 17	576 ± 0	0.00465 ± 0.0023	0.000333 ± 0.00028
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	–	$0.01 \pm 5.1 \times 10^{-7}$	1.41 ± 0.0085	$576 \pm 2.9 \times 10^{-6}$	0.00576 ± 0.0016	$0.000194 \pm 2.8 \times 10^{-11}$
	Separate GP	–	0.017 ± 0.013	7.48 ± 18	592 ± 26	0.00839 ± 0.0071	0.000651 ± 0.00091
	Categorical kernel	–	0.011 ± 0.0054	4.5 ± 12	587 ± 22	0.00608 ± 0.0012	0.000194 ± 0

Table 4: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 10-D	Ackley 2-D	Branin	Golinski	Griewank 2-D	OTL Circuit
LCB	Standard LVGP	0.0439 ± 0.0014	$0.01 \pm 2.6 \times 10^{-8}$	1.41 ± 0.0058	$576 \pm 3.9 \times 10^{-8}$	0.00608 ± 0.0012	$0.000194 \pm 1.4 \times 10^{-10}$
	Heteroscedastic LVGP	0.042 ± 0.00022	$0.01 \pm 4.1 \times 10^{-7}$	27.9 ± 22	$576 \pm 5.2 \times 10^{-10}$	0.0056 ± 0.0018	$0.000194 \pm 2.7 \times 10^{-11}$
	Separate GP	0.0301 ± 0.015	0.017 ± 0.013	22.8 ± 24	576 ± 0	0.00788 ± 0.0068	0.000651 ± 0.00091
	Categorical kernel	0.0456 ± 0.0024	0.0121 ± 0.0075	20.2 ± 22	576 ± 0	0.00712 ± 0.0034	0.000194 ± 0
PI	Standard LVGP	0.0422 ± 0.0016	$0.01 \pm 1.9 \times 10^{-7}$	2.89 ± 8	605 ± 0	$0.0064 \pm 2.5 \times 10^{-6}$	0.00121 ± 0.00064
	Heteroscedastic LVGP	–	0.0115 ± 0.0059	16 ± 22	576 ± 0.0015	0.00736 ± 0.0051	0.000474 ± 0.00034
	Separate GP	0.0309 ± 0.016	0.0161 ± 0.012	21.7 ± 24	576 ± 0	0.00584 ± 0.0043	0.000194 ± 0
	Categorical kernel	0.04 ± 0	$0.01 \pm 4.3 \times 10^{-8}$	11.4 ± 18	576 ± 0	0.00624 ± 0.00086	0.000333 ± 0.00028
EI	Standard LVGP	0.0446 ± 0.0023	$0.01 \pm 1.3 \times 10^{-7}$	1.41 ± 0.0072	602 ± 6.2	$0.0064 \pm 8.6 \times 10^{-10}$	$0.000886 \pm 3.6 \times 10^{-6}$
	Heteroscedastic LVGP	–	$0.01 \pm 9.3 \times 10^{-7}$	20.9 ± 22	$576 \pm 3.9 \times 10^{-6}$	0.00572 ± 0.0033	0.000474 ± 0.00034
	Separate GP	0.0311 ± 0.016	0.014 ± 0.01	25 ± 26	576 ± 0	0.00524 ± 0.0035	0.000194 ± 0
	Categorical kernel	0.04 ± 0	$0.01 \pm 9.3 \times 10^{-8}$	16.8 ± 21	576 ± 0	$0.0064 \pm 5.1 \times 10^{-7}$	$0.000235 \pm 8.1 \times 10^{-5}$
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	–	$0.01 \pm 4.7 \times 10^{-7}$	2.83 ± 7.7	$576 \pm 7.1 \times 10^{-6}$	0.00576 ± 0.0016	0.000474 ± 0.00034
	Separate GP	0.0328 ± 0.017	0.014 ± 0.01	21.8 ± 23	576 ± 0	0.00524 ± 0.0035	0.000194 ± 0
	Categorical kernel	0.04 ± 0	$0.01 \pm 7.3 \times 10^{-8}$	7.1 ± 15	576 ± 0	0.00576 ± 0.0016	0.000413 ± 0.00025
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	–	$0.01 \pm 3.2 \times 10^{-7}$	1.41 ± 0.013	$576 \pm 5.7 \times 10^{-6}$	0.00496 ± 0.0022	0.000375 ± 0.00027
	Separate GP	0.0321 ± 0.026	0.011 ± 0.0054	12.1 ± 22	576 ± 0	0.00428 ± 0.0037	$0.000235 \pm 8.1 \times 10^{-5}$
	Categorical kernel	0.032 ± 0.017	$0.01 \pm 9 \times 10^{-8}$	4.64 ± 9.8	576 ± 0	0.00445 ± 0.0037	0.000377 ± 0.00027
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	0.0479 ± 0	$0.01 \pm 5.5 \times 10^{-7}$	1.41 ± 0.0073	576 ± 0.74	0.00576 ± 0.0016	$0.000194 \pm 1.1 \times 10^{-11}$
	Separate GP	0.049 ± 0.047	0.017 ± 0.013	3.41 ± 9.6	589 ± 25	0.00788 ± 0.0068	0.000651 ± 0.00091
	Categorical kernel	0.0357 ± 0.013	0.0131 ± 0.009	2.34 ± 4.7	576 ± 0	0.00563 ± 0.0033	0.000194 ± 0

Table 5: Final mean-variance regret (RAHBO robustness, $MV = f + \alpha\sigma^2$, $\alpha = 1$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 10-D	Ackley 2-D	Branin	Golinski	Griewank 2-D	
LCB	Standard LVGP	–	0.00641 $\pm$ 0.0063	0.344 $\pm$ 0.54	$3.58 \times 10^{-6} \pm 1 \times 10^{-7}$	0.00381 $\pm$ 0.014	1.53×10^{-1}
	Heteroscedastic LVGP	–	0.0302 ± 0.035	0.793 ± 1.1	$1.71 \times 10^{-7} \pm 1.4 \times 10^{-8}$	0.00794 ± 0.019	4.33×10^{-1}
	Separate GP	–	0.63 ± 1.2	2.09 ± 3.3	0 $\pm$ 0	0.0937 ± 0.11	0
	Categorical kernel	–	0.197 ± 0.71	1.11 ± 3.9	0 $\pm$ 0	0.00409 ± 0.014	3.07×10^{-1}
PI	Standard LVGP	–	0.0127 $\pm$ 0.015	0.571 ± 1.4	48.7 ± 30	0.000738 $\pm$ 0.0011	0
	Heteroscedastic LVGP	–	0.549 ± 1.2	2.53 ± 4.5	0.941 ± 1.9	0.0397 ± 0.075	0
	Separate GP	–	0.121 ± 0.56	0.736 ± 1.5	0 $\pm$ 0	0.0251 ± 0.049	1.74×10^{-1}
	Categorical kernel	–	0.0145 ± 0.012	0.273 $\pm$ 0.46	0 $\pm$ 0	0.000797 ± 0.0019	
EI	Standard LVGP	–	0.0123 $\pm$ 0.012	0.0884 $\pm$ 0.13	24.7 ± 26	$3.12 \times 10^{-7} \pm 3.7 \times 10^{-6}$	
	Heteroscedastic LVGP	–	0.0212 ± 0.024	0.516 ± 1	0.000123 ± 0.00015	0.00649 ± 0.017	
	Separate GP	–	0.223 ± 0.77	1.19 ± 1.7	0 $\pm$ 0	0.0281 ± 0.05	0.000
	Categorical kernel	–	0.0132 ± 0.014	0.332 ± 0.54	0 $\pm$ 0	0.000436 ± 0.00059	
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	–	0.0166 ± 0.016	0.412 ± 0.78	3.75 ± 2.1	0.00432 ± 0.014	
	Separate GP	–	0.219 ± 0.78	0.527 ± 1.1	0 $\pm$ 0	0.0261 ± 0.05	0.000
	Categorical kernel	–	0.0122 $\pm$ 0.012	0.294 $\pm$ 0.54	0 $\pm$ 0	0.000882 $\pm$ 0.0015	
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	–	0.024 ± 0.025	0.424 $\pm$ 0.88	0.307 ± 0.61	0.0127 $\pm$ 0.022	
	Separate GP	–	0.027 ± 0.027	0.523 ± 1.1	0 $\pm$ 0	0.0453 ± 0.063	0.000
	Categorical kernel	–	0.0191 $\pm$ 0.024	0.474 ± 1.1	1.49 ± 3	0.025 ± 0.026	
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	–	0.0264 $\pm$ 0.024	0.00144 $\pm$ 0.0029	9.65 $\pm$ 6.6	0.00789 ± 0.019	6.33×10^{-1}
	Separate GP	–	0.73 ± 1.3	0.915 ± 2.1	72.8 ± 60	0.102 ± 0.13	0
	Categorical kernel	–	0.111 ± 0.56	0.288 ± 1.1	29 ± 47	0.00431 $\pm$ 0.014	1.23×10^{-1}

Table 6: Final mean-variance regret (RAHBO robustness, $MV = f + \alpha\sigma^2$, $\alpha = 1$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 10-D	Ackley 2-D	Branin	Golinski	Griewank 2-D	
LCB	Standard LVGP	1.14 ± 0.23	0.00408 $\pm$ 0.0047	0.00407 $\pm$ 0.0032	$3.48 \times 10^{-6} \pm 1.1 \times 10^{-7}$	0.00381 $\pm$ 0.014	3.87 $\times$
	Heteroscedastic LVGP	0.657 $\pm$ 0.12	0.0188 ± 0.021	1 ± 1.6	$1.72 \times 10^{-7} \pm 1.8 \times 10^{-9}$	0.0097 ± 0.021	4.64 $\times$
	Separate GP	1.38 ± 0.66	0.733 ± 1.3	2.34 ± 3.4	0 $\pm$ 0	0.0784 ± 0.1	
	Categorical kernel	1.18 ± 0.23	0.297 ± 0.88	1.77 ± 4.3	0 $\pm$ 0	0.019 ± 0.064	
PI	Standard LVGP	0.479 ± 0.33	0.0131 ± 0.014	0.139 $\pm$ 0.74	72.5 ± 0	0.000902 $\pm$ 0.0012	
	Heteroscedastic LVGP	–	0.281 ± 1	2.41 ± 4.5	1.37 ± 2.6	0.0436 ± 0.085	
	Separate GP	1.74 ± 0.94	0.715 ± 1.3	1.72 ± 2.2	0 $\pm$ 0	0.0343 ± 0.064	6.94$\times$
	Categorical kernel	0 $\pm$ 0	0.00491 $\pm$ 0.006	0.854 ± 3.3	0 $\pm$ 0	0.00234 ± 0.01	
EI	Standard LVGP	1.48 ± 0.73	0.0102 ± 0.0098	0.00449 $\pm$ 0.0035	62.9 ± 19	$-6.55 \times 10^{-7} \pm 4.2 \times 10^{-7}$	
	Heteroscedastic LVGP	–	0.022 ± 0.031	0.868 ± 1.7	$6.77 \times 10^{-5} \pm 8.2 \times 10^{-5}$	0.0218 ± 0.049	
	Separate GP	1.74 ± 0.92	0.426 ± 1.1	1.81 ± 2	0 $\pm$ 0	0.0278 ± 0.05	3.07$\times$10
	Categorical kernel	0 $\pm$ 0	0.01 $\pm$ 0.0092	0.765 ± 1.3	0 $\pm$ 0	0.000174 ± 0.00025	
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	–	0.0259 ± 0.023	0.17 ± 0.79	7.82 ± 4.1	0.00804 $\pm$ 0.019	
	Separate GP	2.39 ± 0.4	0.429 ± 1.1	1.29 ± 1.8	0 $\pm$ 0	0.0277 ± 0.05	3.07$\times$10
	Categorical kernel	0 $\pm$ 0	0.00801 $\pm$ 0.0085	0.0895 $\pm$ 0.19	0 $\pm$ 0	0.00886 ± 0.02	
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	–	0.0208 ± 0.019	0.00777 $\pm$ 0.022	0.965 ± 1.4	0.018 $\pm$ 0.026	
	Separate GP	3.12 ± 1.3	0.117 ± 0.56	1.42 ± 2.1	0 $\pm$ 0	0.0392 ± 0.049	0.00
	Categorical kernel	2.04 $\pm$ 0.32	0.00552 $\pm$ 0.0094	0.159 ± 0.47	0 $\pm$ 0	0.0387 ± 0.05	
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	1.28 ± 0	0.0203 $\pm$ 0.025	0.00399 $\pm$ 0.012	2.72 $\pm$ 2.4	0.00786 $\pm$ 0.019	2.74 $\times$
	Separate GP	1.73 ± 1.1	0.73 ± 1.3	0.581 ± 1.5	28.8 ± 51	0.0783 ± 0.1	
	Categorical kernel	0.574 $\pm$ 0.37	0.4 ± 1	0.717 ± 3.1	10.6 ± 4	0.0403 ± 0.1	